Bulk-Silicon Parent Qualification and Joint-Reduction Technical Report

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Draft calculation report

Abstract

This report defines the calculation boundary for a scalar-relativistic, non-spin-polarized, non-spin–orbit-coupled bulk-silicon parent, a compatible ten-spatial-orbital Wannier bridge, and a finite-support local-model hierarchy. Current calculation-owned evidence includes an accepted right-handed relaxed structure/settings pair for the retained SSSP comparison branch, an authenticated plane-wave-to-Wannier interface, one provider-eligible rank-10 Wannier occurrence, and a prospectively frozen 408-point direct-parent comparison. The valley-position target was met, but the indirect-gap and effective-mass targets were not; the candidate is therefore not adequate for the declared SSSP-pilot intended use. The report also includes an authorized producer-checked Hamiltonian gauge-alignment/locality development result, frozen Wannier-shell fidelity studies, and a comprehensive non-SOC band-edge amendment. Component-operator and material-validation claims remain pending.

1 Evidence status

This is a publication-independent calculation report. It distinguishes the historical provisional bootstrap from the current right-handed SSSP comparison branch and from any future primary pseudopotential branch. The current branch has a provider-eligible Wannier occurrence whose source reconstruction, descriptive characterization, and withheld intended-use comparison are complete. It is not an accepted Wannier operator: the frozen comparison found a 14.50% indirect-gap relative error, two failed longitudinal-mass comparisons, and failed transverse-mass comparisons at every valley. The six valley-position comparisons met their target. No completed component-operator construction, model-selection result, or broader scientific validation follows from this result. Exact calculator occurrences and their authorizations are retained outside this report and referenced by calculation-owned evidence records.

2 Scientific objective

The target is a pristine bulk-silicon parent represented consistently at three levels:

1. a first-principles plane-wave oracle $O_A(\mathbf{k})$;
2. a retained ten-spatial-orbital Wannier bridge $O_B(\mathbf{k})$; and
3. a finite-support sp^3s^* local-model hierarchy $O_C^{U,S}(\mathbf{k})$.

For retained frame $Q(\mathbf{k})$, alignment $U(\mathbf{k})$, support S, and real-space blocks $T_U(\mathbf{R})$, the intended relations are

$$O_B(\mathbf{k}) = Q(\mathbf{k})^\dagger O_A(\mathbf{k}) Q(\mathbf{k}), \tag{1}$$

$$O_C^{U,S}(\mathbf{k}) = \sum_{\mathbf{R}\in S} e^{i\mathbf{k}\cdot\mathbf{R}} T_U(\mathbf{R}). \tag{2}$$

A candidate is adequate only if the same model, support, alignment, basis, and energy reference satisfy both declared spectral and aligned-operator tolerances. Visual band agreement alone is insufficient.

3 External empirical development control

The calculation-local Vogl sp^3s^* control follows the empirical model of Vogl, Hjalmarson, and Dow [1] and contains the accepted complete Table 1 transcription: 16 compounds, 13 parameters per compound, and energies in eV. An independent implementation constructs the nearest-neighbor, spinless, non-SOC ten-band Bloch Hamiltonian in the ordered anion–cation basis $(s, p_x, p_y, p_z, s^*)_a \oplus (s, p_x, p_y, p_z, s^*)_c$. Its reduced Cartesian wavevector uses $\mathbf{k} = (2\pi/a_L)\mathbf{q}$.

Software and analytic checks cover the table inventory, silicon parameter row, structure factors at symmetry points, exact Hermitian assembly, the silicon Γ spectrum, reciprocal periodicity, and invalid-input rejection. This external empirical control is not derived from the provisional first-principles bootstrap or a Wannier bridge and does not freeze the candidate inventory, establish model adequacy, or qualify a material result.

4 Existing evidence

The historical direct bootstrap evaluated six wavefunction-cutoff candidates and four reciprocal-mesh candidates, with exact reuse of the C48 branch as logical K8. This produced nine unique self-consistent-field (SCF) and nine linked diagnostic non-self-consistent-field (NSCF) executions. The compact audit recomputed the required identities and observables and found the retained matrix complete and internally consistent.

Several adjacent comparisons meet the frozen finite-setting criteria. Human disposition selected C48/K8—48/192 Ry with a shifted $8 \times 8 \times 8$ mesh—as a provisional finite-setting parameter pair. This establishes neither an accepted production setting nor an infinite-basis error bound. The provisional geometry is not an accepted equilibrium lattice, the three diagnostic momenta do not locate the conduction valley, and printed fixed-point bands do not converge an effective mass.

Every invocation also emitted the same IEEE flag report. Instrumented diagnoses localized one event to external OpenCL discovery during startup and classified a separate cross-BLAS, cross-MPI divide-by-zero below `ZHEGVX` during initial-wavefunction subspace diagonalization. The effect on retained observables and the sources of the remaining reported flag classes are unresolved. Successful exit and `JOB DONE.` do not establish harmlessness. A mixed cutoff–mesh interaction check implied by C48/K8 would be new protected execution.

Exact values, hashes, criteria, and warning investigations remain at the historical locations catalogued in `calculations/Si/bulk/evidence-crosswalk.md`. They are referenced rather than copied or relabelled.

4.1 Current right-handed SSSP comparison branch

The current branch retained the exact relaxed structure `Si.PrimitiveUnitCell.RightHanded.RelaxedS1`, a Gamma-centered K10 mesh, and 42/336 Ry wavefunction/charge-density cutoffs under calculation-owned finite-setting rules. Separate authenticated SCF, Davidson NSCF, Wannier90 preprocessing, converter, and localization occurrences preserve their own identities and dispositions.

The accepted interface contains 64 ordered k points, 12 non-spin bands, a rank-10 sp^3s^* projection inventory, and byte-authenticated NNKP, AMN, MMN, and EIG artifacts. An initial construction that froze all eight conduction bands but no valence bands remained nonstationary. A lower-energy frozen window $[-5.85, 8.00]$ eV instead freezes all four valence states and zero to two sampled low-energy conduction states per k point. With a fresh 2000-step disentanglement budget, that construction converged disentanglement at iteration 1290 and Wannierisation at iteration 189. It is provider-eligible; no downstream scientific acceptance is implied.

5 Selected C6-B20-W21 gauge-alignment/locality deployment

An authorized Hamiltonian-only development calculation was completed for the authenticated, provider-eligible, rank-10 C_6-B20-W21 represented operator. It applies one frozen rank-10, depth-2 alternating Givens attack with axis harmonics, axis-maximum normalization, and coefficient $\pi/8$. The complete represented spectra remain invariant under the generating unitary to a maximum observed difference of 3.20×10^{-14} eV, while restricted alignment residuals, finite Fourier-block power, and finite-support cost may change. This is a representation study inside the admitted finite Wannier Hamiltonian. It does not reverse the negative revealed-parent gap and effective-mass disposition.

The construction field uses the Gamma-origin $6 \times 6 \times 6$ inventory reindexed to the centered axis $(-1/2, -1/3, -1/6, 0, 1/6, 1/3)$. The evaluation field uses the ordered 408-point revealed development geometry with uniform weight $1/408$. Centered finite-cyclic blocks and native Wannier90 HR/Wigner–Seitz blocks remain distinct routes. Ordinary conversion retains the 125 displacements with components in $\{-2, -1, 0, 1, 2\}$. For lattice parameter $a = 5.469182093640207$ Å, the shell scale is $4/a^2$, the integer shell label is

$$q = 2\left(n_1^2 + n_2^2 + n_3^2 + n_1n_2 + n_1n_3 + n_2n_3\right), \tag{3}$$

and the complete ordinary inventory ends at $q = 48$.

No Hamiltonian or band acceptance threshold is introduced. The report instead retains all complete support-error curves, Pareto comparisons, and equal-attained-error support projections. Errors are reported in eV and through a declared 1 eV normalization scale. The numerical figures may consume the authenticated producer result directly. They must carry the status `producer-checked-development-analysis-not-independently-verified`; no independent reconstruction is required for this development version. Identity and global-unitary alignment retain maximum frame defects of approximately 1.021 and 1.020, whereas the pointwise-polar and known-inverse controls reduce that defect to approximately 1.29×10^{-15} and 5.06×10^{-16}. The reference and identity-attacked non-onsite powers are 20.7574 and 21.5057 eV^2 per retained state. The finite-cyclic interpolation maximum sorted-band errors are 0.3470 and 0.3867 eV; these conversion/interpolation consequences remain separate from the complete-field gauge-invariance residual.

5.1 Retained figure package

The complete figure package is part of the technical-report contract. Reference and attacked channels must remain distinguishable without color, zeros on logarithmic axes must use explicit below-floor markers, and no curve may be smoothed across unsupported shells or reciprocal points. Interpolation, deliberate truncation, represented total, and direct-parent overlay channels must not be stacked as if they were additive.

6 Post-hoc two-tolerance Wannier-shell exploration

The central two-tolerance question is explored post hoc without a Slater–Koster parameterization. The candidate family is instead the exact frozen hierarchy of native complete Wigner–Seitz shell truncations in the same authenticated rank-10 Wannier basis. No fitted alignment is required. For candidate shell j, define

$$\tau_{1,j}^{\mathrm{req}} = \left[\frac{1}{4080} \sum_{\mathbf{k},n} \left(\frac{\epsilon_{n\mathbf{k}}^{(j)} - \epsilon_{n\mathbf{k}}^{\mathrm{W}}}{1\,\mathrm{eV}} \right)^2 \right]^{1/2}, \tag{4}$$

$$\tau_{2,j}^{\mathrm{req}} = \left[\frac{1}{4080} \sum_{\mathbf{k}} \frac{\| H_j(\mathbf{k}) - H_{\mathrm{W}}(\mathbf{k}) \|_{\mathrm{F}}^2}{(1\,\mathrm{eV})^2} \right]^{1/2}. \tag{5}$$

The ordered 408-point development inventory has uniform weights. A post-hoc threshold pair (τ_1, τ_2) is feasible when at least one proper reduced shell satisfies $\tau_{1,j}^{\mathrm{req}} \le \tau_1$ and $\tau_{2,j}^{\mathrm{req}} \le \tau_2$. The displayed witness is the feasible shell with the smallest exact representative count.

The exact full-support source, H-shell-17 with 383 representatives, is retained as the zero-loss control but excluded from reduced-family feasibility; otherwise every nonnegative pair would be trivially feasible. Among the 17 proper reductions, H-shell-16 minimizes both primary losses while retaining 371 representatives:

$$(\tau_{1,16}^{\mathrm{req}}, \tau_{2,16}^{\mathrm{req}}) = (4.4635917 \times 10^{-4}, 1.5173254 \times 10^{-3}). \tag{6}$$

Thus this literal shell family shows no global spectral–operator conflict at its tightest proper reduction: the same H-shell-16 candidate minimizes both primary losses. Smaller supports retain local nonmonotonic spectral/operator tradeoffs, so the nontrivial post-hoc object is the single minimum-support witness surface across the threshold plane. For example, $(0.1, 0.15)$ first admits H-shell-07 with 135 representatives, $(0.01, 0.02)$ first admits H-shell-13 with 313, and $(0.001, 0.002)$ first admits H-shell-16 with 371. The pair $(0.0004, 0.001)$ admits no proper reduced shell. These examples are descriptive readings of the completed map, not prospectively selected acceptance thresholds.

6.1 Refitted native-shell follow-on

The near-linearity of the literal requirement points is expected because nested shell deletion changes only one effective coordinate and both losses are driven by the same omitted operator. A separately authorized post-hoc follow-on therefore retained the same Wannier basis and native complete supports but introduced one real scale per retained shell,

$$H_{\boldsymbol{\alpha}}(R) = \alpha_s H_{\mathrm{W}}(R), \qquad 0 \le \alpha_s \le 2, \tag{7}$$

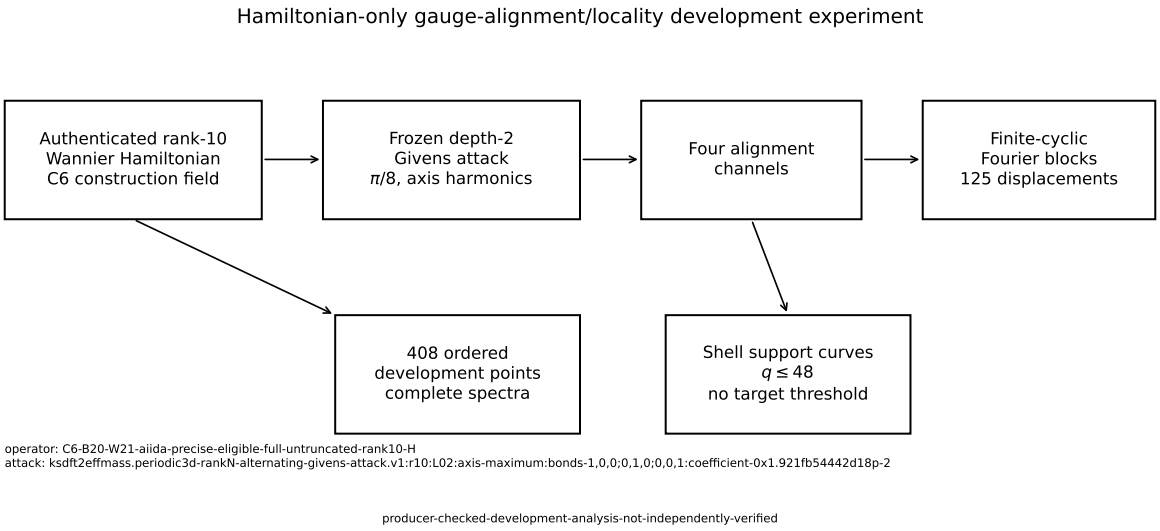


Figure 1: Hamiltonian-only route from the authenticated native represented operator through the frozen gauge attack, alignment channels, centered finite transform, ordinary conversion, complete block supports, and producer-derived report projections. All numerical values are producer-checked development outputs and are not independently reconstructed.

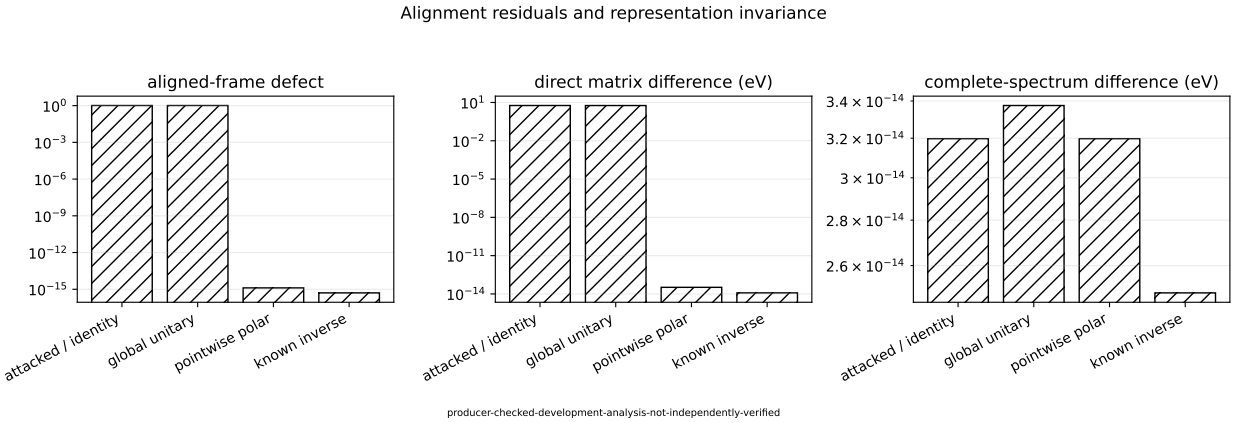


Figure 2: Identity, one-global-unitary, pointwise-polar, and known-inverse channels, showing aligned-frame defects, direct matrix residuals, and complete ordered-spectrum residuals. Pointwise and known-inverse channels are oracle controls. All numerical values are producer-checked development outputs and are not independently reconstructed.

with one common α_s for every effective representative in shell s, including R and $-R$. Blocks outside the declared support remain zero. This preserves the native within-shell block patterns and introduces neither atomic-orbital labels nor a Slater–Koster parameterization.

The producer searched 17 spectral/operator scalarizations from three deterministic starts using bounded L-BFGS-B and analytic binary64 gradients. The resulting converged constructive witnesses form distinct spectral/operator tradeoff arcs rather than the literal deletion line. At the illustrative pair $(0.1, 0.15)$, the minimum observed support falls from literal H-shell-07 with 135 representatives to fitted H-shell-05 with 79. At $(0.05, 0.1)$ it falls from 177 to 135 representatives, and at $(0.02, 0.04)$ from 225 to 201. The tighter pairs $(0.01, 0.02)$, $(0.002, 0.006)$, and $(0.001, 0.002)$ retain the same minimum observed support counts as the literal study.

Within H-shell-16 itself, an operator-oriented converged witness attains approximately $(4.04 \times 10^{-4}, 1.394 \times 10^{-3})$, whereas a spectral-oriented witness attains approximately $(2.98 \times 10^{-4}, 1.745 \times 10^{-3})$. This is a same-support spectral/operator tradeoff produced by refitting retained native blocks rather than by changing the Wannier basis.

6.2 Wannier-only fidelity ladder

Three fitted witnesses were then frozen without further optimization: H-shell-05 with 79 representatives, H-shell-10 with 201, and H-shell-16 with 371. Their operator RMS errors over the normalized Brillouin torus were calculated exactly from the unique integer Fourier coefficients using Parseval's identity. Spectral errors were evaluated independently on Gamma-origin 12^3, 20^3, and 32^3 meshes, followed by a source-relative six-valley, indirect-gap, Hessian, and principal-mass analysis.

The fitted 408-point inventory materially understates broader error. For H-shell-05, the spectral RMS rises from 0.0724 to 0.1820 eV and the operator RMS from 0.1281 to 0.2600 eV. The corresponding H-shell-10 changes are 0.0132 to 0.0279 eV and 0.0323 to 0.0596 eV. For H-shell-16 they are 4.04×10^{-4} to 6.57×10^{-4} eV and 1.394×10^{-3} to 1.747×10^{-3} eV. The sampled operator values agree with the exact Parseval values to at most 5.56×10^{-17} eV on all three meshes, while the 12^3–32^3 spectral RMS changes remain below 1.4% in magnitude.

Feature fidelity is more demanding than global spectral RMS. Relative to the full Wannier source, H-shell-05 changes the gap by -65.8 meV, displaces a valley by as much as 0.0948 Å^{-1}, and changes at least one principal mass by 67.7%. H-shell-10 changes the gap by -34.1 meV and a principal mass by 53.2% despite its 32^3 all-band spectral RMS being below 0.028 eV. H-shell-16 changes the gap by -0.503 meV, the maximum valley location by 0.00276 Å^{-1}, and the maximum principal mass by 2.54%.

6.3 Comprehensive non-SOC band-edge amendment

The fidelity ladder's initial feature layer was conduction-centered. A subsequent no-calculator amendment retained the same four frozen Hamiltonians and added complete Cartesian electron mass and inverse-mass tensors, finite-difference sensitivity, derived six-valley density-of-states and equal-population conductivity-mass conventions, Gamma-valence cluster integrity, high-symmetry direct gaps, ordered-band bandwidths, and a degeneracy-aware valence-subspace fit. The derived mass conventions summarize represented curvatures only; they are not finite-temperature DOS or transport results.

The full source's three-band Gamma-valence splitting is approximately 0.02495 meV. H-shell-05, H-shell-10, and H-shell-16 increase it to approximately 102.2, 54.8, and 2.50 meV. Their maximum principal angles from the source cluster are only 1.856°, 0.925°, and 0.0420°, respectively. Thus a

reduced model may preserve the valence subspace while substantially damaging the Hamiltonian inside that subspace.

Independently ordered valence eigenvalues do not define stable individual-band Hessians at this near-degeneracy. In the full source, the upper ordered branch is not hole-like for every tested fit window in any of the three directions, and finite ordered-branch masses vary by as much as approximately 75.5%. The apparently stable H-shell-05 and H-shell-10 ordered branches are not evidence of superior hole physics: their large artificial Gamma splittings have removed the source degeneracy.

The primary valence result instead polar-transports each displaced three-band cluster to its Gamma cluster and fits the even represented matrix field to

$$H_{\mathrm{eff}}(\mathbf{q}) = H_{\mathrm{eff}}(0) + q_x^2 B_{xx} + q_y^2 B_{yy} + q_z^2 B_{zz} \tag{8}$$

$$+ 2q_x q_y B_{xy} + 2q_x q_z B_{xz} + 2q_y q_z B_{yz}. \tag{9}$$

Directional curvature channels are the eigenvalues of twice the direction-contracted quadratic coefficient matrix. At the 0.01 Å^{-1} fit window, the full-source apparent channel masses are $(0.141, 0.277, 0.301)$ along [100], $(0.0969, 0.312, 2.923)$ along [110], and $(0.0877, 0.697, 0.772)$ along [111], in electron mass units. These are eigenchannels of a scalar-relativistic non-SOC three-band quadratic operator, not heavy-hole, light-hole, or split-off masses.

After cluster alignment, H-shell-05, H-shell-10, and H-shell-16 have joint quadratic coefficient errors of approximately 9.54%, 5.88%, and 0.279%. Their maximum curvature-channel mass changes are approximately 38.9%, 19.7%, and 4.53%. The matrix fits are stable across the tested windows: the joint coefficient field changes by at most approximately 0.41%, and primary fit residuals are approximately 5–6 $\times 10^{-7}$ eV RMS.

The source six-valley DOS-mass convention is $1.4764\, m_e$, and its equal-population conductivity-mass convention is $0.25063\, m_e$. H-shell-05 changes these by approximately 16.9% and 20.4%; H-shell-10 by 22.9% and 6.90%; and H-shell-16 by 0.333% and 0.0216%. The maximum principal-mass spread across the four finite-difference steps is below approximately 0.172% for every model, so the larger source-relative changes are not controlled by the selected stencil.

The source equivalent-point direct-gap spreads are approximately 0.00366 meV at X and 0.0239 meV at L. The corresponding H-shell-05 spreads are 150.1 and 68.7 meV; H-shell-10 gives 36.7 and 63.9 meV; and H-shell-16 gives 0.427 and 2.97 meV. The moderate reductions therefore amplify residual cubic-equivalence violations. The maximum ordered-band bandwidth changes on the independent 32^3 mesh are approximately 354, 81.1, and 3.14 meV for H-shell-05, H-shell-10, and H-shell-16.

These native-shell studies test internal Wannier-shell compressibility only. They are not prospective compatibility decisions for the originally proposed Slater–Koster class, DFT-parent validations, or evidence that thresholds chosen after inspecting the maps are scientific acceptance limits. The fitted search supplies no global infeasibility certificate, and the fidelity ladder does not repair the source's failed DFT-parent gap and effective-mass criteria.

7 Required parent qualification

The production parent must freeze the physical branch, lattice, PBE/ONCV approximation, Quantum ESPRESSO executable, pseudopotential identity, occupations, solver settings, cutoff and mesh inventory, compact/native artifact boundary, and warning policy before execution. Qualification must keep energy, stress, fixed-point bands, gaps, valley identity, and curvature requirements distinct.

For each retained observable q, finite-setting agreement

$$|q_j - q_{j+1}| \le \tau_q \tag{10}$$

demonstrates only stability across the tested pair. It does not imply a bound on $|q_j - q_\infty|$ without an additional reviewed error argument.

8 Required retained-space bridge

The Wannier bridge must retain projections, windows, centers, spreads, convergence status, reciprocal sewing, symmetry and orbital semantics, energy reference, and direct parent comparisons. At crossings or degeneracies, comparison must use spectral clusters or projectors rather than undocumented individual eigenvector labels.

Wannier90 may produce a candidate witness. Wannier90 convergence is not by itself a proof of the bridge contract. The tutorial-scale silicon Wannier extraction remains an integration precursor and is excluded from the production admissible sets.

9 Optimization diagnosis and global-minimum nonclaim

Wannier construction, local optimization, arithmetic stability, and scientific adequacy are separate questions. Rank, bands, projections, windows, reciprocal mesh, and neighbor topology define the represented target space and initial gauge. Changing one of those coordinates creates a new scientific construction rather than a mere optimizer retry.

Wannier90 performs projection-initialized local spread minimization; it does not certify a global minimum. Iteration exhaustion or large late spread excursions establish nonstationarity or optimizer sensitivity, but do not prove that an acceptable gauge is absent. Conversely, arithmetic or toolchain instability requires direct evidence such as non-finite data, corrupted identities, failed matrix invariants, same-input nondeterminism, or controlled sensitivity to the arithmetic implementation. Optimizer excursions alone are insufficient.

The all-conduction-frozen construction exhibited large late excursions through iteration 3934 and material between-mesh gauge sensitivity. The lower-energy frozen construction converged using the same executable and byte-identical interface artifacts, with gauge-unitarity and disentanglement-isometry residuals of approximately 1.58×10^{-10} and 1.32×10^{-10} and a zero reported Hamiltonian Hermiticity residual. The supported diagnosis is therefore construction-induced optimization-landscape sensitivity, not floating-point failure. This diagnosis does not prove that the converged occurrence is a global spread minimum.

A certified global minimum is not required for acceptance. Instead, a locally converged occurrence must be structurally valid and must satisfy prospectively declared construction- and withheld-mesh spectral, valley, effective-mass, and locality tests. Competing-minimum studies become mandatory when distinct converged initializations produce materially different target observables or locality conclusions. The detailed decision procedure is maintained in `calculations/Si/bulk/research-protocol/research-protocol.tex`.

10 Joint spectral/operator decision

For a frozen finite candidate inventory D, define

$$F(\tau_E, \tau_O) = \{d \in D : L_E(d) \le \tau_E \ \wedge \ L_O(d) \le \tau_O\}\,. \tag{11}$$

If this set is nonempty, the intended finite decision is

$$C^*(\tau_E, \tau_O) = \min_{d \in F(\tau_E, \tau_O)} C(d). \tag{12}$$

The same d must determine the model, support, alignment, and energy reference in both loss channels. Separate spectral and operator optima do not form a common witness.

The decision must preserve distinct numerical, subspace, alignment, interpolation, and support-truncation error channels. It must report one of the following:

- a common witness and the smallest adequate declared model;
- a reviewed separation result for an exhaustively evaluated finite inventory; or
- an explicit inconclusive outcome.

Failure of a local optimizer does not prove infeasibility.

11 Withheld analysis and residual attribution

Construction and withheld momentum sets must be disjoint by contract. Withheld analysis must report band, gap, valley, effective-mass, projector, represented-operator, and locality channels separately. Stable residuals may be attributed by onsite or hopping block, orbital semantics, symmetry channel, and support shell only when the representation and alignment contracts permit that interpretation.

Sensitivity analysis must include every prospectively declared parent, Wannier, stencil, alignment-map resolution, and threshold variation. Post-hoc sensitivity does not retroactively justify thresholds.

12 Verification and retention

Publication-qualified parent and model-selection calculations require independent verification of sources, schemas, identities, finite candidate coverage, objective reconstruction, threshold evaluation, and withheld results. The selected C6-B20-W21 gauge-alignment/locality deployment is an explicit development exception: its plots may consume authenticated producer records directly and must remain labeled as not independently reconstructed. A retained package must be no-overwrite and checksum-bound.

Formal proofs or certificates may establish exact implications for finite definitions and accepted interval premises. They do not certify Quantum ESPRESSO or Wannier90 execution, JSON, binary64 rounding, pseudopotential transferability, or physical validation.

13 Current gate and next decision

The lower-energy frozen-window occurrence remains provider-eligible and structurally valid, but its intended-use comparison is now complete and negative. One authorized 408-point Davidson NSCF completed without an unconverged-eigenvalue diagnostic. All six candidate-to-parent valley displacements, ranging from 0.02120 to 0.03086 Å^{-1}, met the frozen 0.034465 Å^{-1} target. The candidate indirect gap was 0.52261 eV versus the direct-parent 0.61125 eV, a 14.50% relative error against the 5% target. On the prospectively frozen candidate-centered primary stencils, candidate transverse masses of approximately 0.262–0.304 m_e disagreed with direct-parent local-curvature

values of approximately 0.193–0.208 m_e by 32–56%, and two longitudinal comparisons also exceeded the 8% target.

The research protocol now recommends a one-variable successor C_6: change only the Gamma-origin construction mesh from $4 \times 4 \times 4$ to $6 \times 6 \times 6$, while holding the accepted structure and parent, 12-band inventory, rank 10, windows, projections, and optimizer controls fixed. The revealed 408-point parent set becomes development evidence for successors. A separate 402-point parent-informed, rotated three-dimensional confirmation geometry is frozen with its direct-parent values still unobserved. A C_8 mesh is conditional on review of the channel-resolved C_4-to-C_6 changes and is not automatic. A C_{12} construction with 1728 points is reserved as a bounded upper diagnostic only when a completed C_8 review continues to support the mesh-density hypothesis. Passing at C_{12} would establish adequacy of that finite operator for the declared use, not an infinite-mesh result.

The exact C_6 NSCF input specified 216 points, 12 bands, one process, zero retries, a 600-second timeout, and a 512 MiB retention ceiling. A later exact authorization permitted one invocation. QE completed in 28.04 seconds wall time with finite 216-point/12-band QEXSD, no unconverged-eigenvalue warning, and a verified 219-file saved state. Under the unchanged outer window, 142 points retain all 12 parent bands, 62 retain 11, and 12 retain the rank-minimum 10; the frozen window contains four to six states per point. This effective-inventory change is a consequence of refining the mesh while holding the numerical windows fixed, not an independently adjusted window.

The NSCF is an eligible handoff, not a Wannier operator. One later bounded `wannier90.x -pp` authorization produced a source-correlated 216-point NNKP with eight neighbors per point, 1728 ordered neighbor records, ten expanded projections, and no excluded bands. One separately authorized `pw2wannier90.x` occurrence then produced finite AMN/MMN/EIG artifacts with the declared 216-point, 12-band, rank-10 shapes, exact NNKP-neighbor correlation, and maximum EIG-to-QEXSD discrepancy 5.23×10^{-13} eV. One separately authorized exact localization occurrence then completed natively but failed provider eligibility: after the fixed 2000 Disentanglement iterations, the absolute fractional change was 4.679×10^{-8}, 467.9 times the unchanged dimensionless 10^{-10} tolerance. Wannierisation later converged at iteration 316, but cannot repair the unconverged subspace. Thus no provider-eligible C_6 operator exists. A calculation-owned amendment then permitted one fresh occurrence changing only `dis_num_iter` from 2000 to 4000 and consuming no failed checkpoint. Disentanglement converged at iteration 3068, but Wannierisation did not satisfy its six-consecutive-step gate by iteration 4000. The final absolute spread change, 1.26×10^{-5} Å^2, was 6300 times the unchanged criterion, and large late excursions remained. The provider rejected the output and the prospective stop rule forbids automatic budget extension or C_8 escalation.

A subsequent parent-band review selected $C_{6\text{-}B20}$ as the next staged candidate. It adopts only the moderate band-count heuristic from the pinned `aiida-wannier90-workflows` 3.1.0 implementation [2]. That workflow uses `num_bands_factor`=2.0 and chooses the maximum of scaled occupied-state and projection counts and their plus-four safeguards. For eight Si valence electrons, ten declared projections, and a non-spin-polarized calculation, the calculation-owned adaptation gives $\max(8, 8, 20, 14) = 20$ bands. This citation does not adopt the workflow's default PBEsol pseudopotential or its complete automation protocol. One separately authorized NSCF changed only the parent inventory from 12 to 20 bands; rank 10 remained fixed. It completed successfully with 216 points, finite values, and a verified saved state. The inherited $[-5.85, 19.02]$ eV window contains 10–14 states per point, including 68 newly observed states from bands 13 and 14. The prepared $[-6.0, 21.0]$ eV window contains 12–14 states, including 228 states from bands 13 and 14; band 15 remains above 21 eV everywhere. The subsequent target-space decision selects $C_{6\text{-}B20\text{-}W21}$ with outer window $[-6.0, 21.0]$ eV. Its upper boundary lies 0.1964 eV above the observed band-12 maximum and 0.2572 eV below the band-15 minimum, providing at least two outer-space dimensions beyond rank 10 at every construction point. This is an eligibility design, not evidence of optimizer

convergence or scientific adequacy.

The prospective route now also adopts the pinned AiiDA precise convergence-derived tuple for the exact two-atom cell: `conv_tol` and dimensionless `dis_conv_tol` are each 2.0×10^{-9} after site-count scaling, `conv_window`=6, `num_cg_steps`=200, and both iteration ceilings are 40000. This coherent tuple supersedes the unexecuted hybrid 4000-step preflight. It does not adopt the precise factor-three band count, automatic mesh rule, PBEsol pseudopotential, projections, energy windows, or workflow retry machinery. One separately authorized fresh preprocessing occurrence returned zero and produced an eligible, authenticated 216-point NNKP with the declared rank-10 projection inventory. Its terminal `IEEE_OVERFLOW_FLAG` line is retained as a nonblocking diagnostic because native completion, bounded parsing, and every declared NNKP invariant passed; preprocessing does not exercise the localization optimizer. The existing 12-band EIG, AMN, and MMN files still cannot represent bands 13–20. One separately authorized conversion occurrence then returned zero and produced finite, shape-correct 20-band AMN/MMN/EIG artifacts. The MMN neighbor inventory matches the NNKP exactly, and the maximum EIG-to-QEXSD discrepancy is 5.37×10^{-13} eV. The terminal QE IEEE flag line remains a nonblocking diagnostic under the same native-completion, finiteness, and correlation rule. A fresh localization preflight bound the exact interface to the 40000-step precise profile without any checkpoint or failed-output consumption. One separately authorized occurrence converged natively at disentanglement iteration 1517 and Wannierisation iteration 378, with total spread 21.022751219 Å^2. An initial provider rejection was a false negative caused by comparing the ten-decimal matrix coordinates with the lossy eight-decimal NNKP presentation. ProjectKoios was corrected to use the canonical WIN k-point block. No-calculator reverification then measured gauge and disentanglement residuals of 3.30×10^{-11}, passing the unchanged 10^{-10} fractional gate. The occurrence is provider-eligible. A separately authorized no-calculator S30 analysis then characterized the full untruncated operator. It verified all real-space pairs with zero Hermiticity residual, reproduced 954 frozen states within 1.29×10^{-9} eV, located all six declared Delta-family minima, found positive principal curvatures, and retained the complete 383-representative locality curve. The initial fixed C4-era reconstruction gates were preserved as failed diagnostics and replaced for structural verification by conservative bounds derived from the authenticated six-decimal HR, ten-decimal transformation matrices, twelve-decimal EIG values, exact degeneracies, dimensions, and energy scale. No spatial truncation was applied. Revealed-408 development comparison is next; confirmation, spatial truncation, and Slater–Koster reduction remain blocked. The factor-three 30-band adaptation remains a conditional upper diagnostic rather than an automatic successor. The prepared $C_{6\text{-}T1}$ localization was never executed and is not the selected next occurrence.

Component-operator construction and a Slater–Koster reduction remain blocked until a full Wannier operator meets its prospectively declared direct-parent criteria. The local projection inventory does not itself impose Slater–Koster shells or parameters. Detailed current gates and evidence remain in `calculations/Si/bulk/simulations/` and `calculations/Si/bulk/development-plan.md`.

14 Nonclaims

This report does not establish production convergence, an infinite-basis result, experimental agreement, pseudopotential transferability, a qualified Wannier model, an adequate tight-binding model, or scientific validation. It authorizes no Quantum ESPRESSO or Wannier90 execution, evidence promotion, publication, or release.

A Supplemental material

A.1 Technical note: Vogl-control implementation provenance

The scientific definition and parameter authority for the empirical control are the published Vogl model and its Table 1 [1]. A separate development lineage was inspected for comparison: a public zinc-blende LCAO Gist by Mahdisadjadi [3] and the 2026 `SiliconLCAO_4sDOS-v3` notebook attributed within the notebook to Simon Phillpot [4]. The notebook states that it began from the Gist, corrected phase factors, and added the s^* orbitals, with modifications dated 29 April through 11 May 2026.

The calculation-local module `vogl1983_lcao_sp3sstar/model.py` was written independently from the published model equations and the accepted Table 1 transcription. No Gist or notebook source code was copied. Their role is limited to development provenance and numerical cross-checking; they do not supersede the primary source. The notebook agreed with 203 of the 208 Table 1 cells at four-decimal numerical normalization. The five disagreements were resolved in favor of the printed Vogl values. Neither the Gist nor the notebook exposed an explicit license in the inspected material, which is an additional reason to keep their role separate from the maintained implementation.

A.2 Technical note: LCAO and tetrahedral sp^3 construction

This summary follows the tight-binding treatments of Harrison, Chadi and Cohen, and Vogl et al. [5, 6, 1]. At one site, begin with the orthonormal orbitals $|s\rangle$, $|p_x\rangle$, $|p_y\rangle$, and $|p_z\rangle$,

$$\langle \alpha \mid \beta \rangle = \delta_{\alpha\beta}, \qquad \alpha, \beta \in \{s, p_x, p_y, p_z\}. \tag{13}$$

The four sp^3 hybrid orbitals are formed by combining the s, p_x, p_y, and p_z orbitals with equal magnitudes but different sign patterns:

$$\begin{pmatrix} |h_1\rangle \\ |h_2\rangle \\ |h_3\rangle \\ |h_4\rangle \end{pmatrix} = \frac{1}{2} \underbrace{\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}}_{H} \begin{pmatrix} |s\rangle \\ |p_x\rangle \\ |p_y\rangle \\ |p_z\rangle \end{pmatrix}. \tag{14}$$

The sign matrix H is a 4×4 Hadamard matrix. Its rows are mutually orthogonal and satisfy

$$H H^{\mathsf{T}} = 4I. \tag{15}$$

Consequently, $H/2$ defines an orthogonal change of basis that preserves norms and inner products. It transforms the orthonormal atomic-orbital basis into four orthonormal hybrid orbitals, $\langle h_m \mid h_n \rangle = \delta_{mn}$. Each row's sign pattern selects one of the $(1,1,1)$, $(1,\bar{1},\bar{1})$, $(\bar{1},1,\bar{1})$, or $(\bar{1},\bar{1},1)$ tetrahedral bond directions. Because this normalized Hadamard transform is symmetric and orthogonal, the same matrix also recovers the original s and p orbitals from the hybrid basis.

For a periodic solid, form Bloch sums of localized orbitals,

$$|\alpha b \mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\nu}_b)} |\alpha b \mathbf{R}\rangle, \tag{16}$$

where b labels the anion or cation sublattice. Translation therefore enters a Hamiltonian matrix element through a phase $e^{i\mathbf{k}\cdot\mathbf{r}}$; reversing the bond gives the complex-conjugate phase required by

Hermiticity. The LCAO approximation represents each symmetry-distinct localized matrix element by a constant: onsite E_s and E_p, and nearest-neighbor V_{ss}, V_{sp}, V_{xx}, and V_{xy}. In a heteropolar crystal, anion and cation onsite energies differ, and $V(s_a, p_c)$ and $V(s_c, p_a)$ are separate directional parameters. Onsite s–p matrix elements vanish by the site symmetry of the orthogonal basis; intersite s–p entries acquire signs from the odd parity of the p orbital.

For the four nearest-neighbor displacements

$$\mathbf{r}_{1,2,3,4} = \frac{a_L}{4}\big\{(1,1,1),(1,-1,-1),(-1,1,-1),(-1,-1,1)\big\}, \tag{17}$$

write $e_n = e^{i\mathbf{k}\cdot\mathbf{r}_n}$. The four structure factors are

$$g_0 = \tfrac{1}{4}(e_1 + e_2 + e_3 + e_4), \qquad g_1 = \tfrac{1}{4}(e_1 + e_2 - e_3 - e_4), \tag{18}$$

$$g_2 = \tfrac{1}{4}(e_1 - e_2 + e_3 - e_4), \qquad g_3 = \tfrac{1}{4}(e_1 - e_2 - e_3 + e_4). \tag{19}$$

Thus representative Bloch-matrix entries are $g_0 V(s,s)$ for s–s coupling and $g_j V(s_a, p_{j,c})$ for s–p_j coupling, with the reverse entries fixed by the Hermitian conjugate and the orbital-direction sign convention. The sp^3 construction produces an 8×8 Hamiltonian for two four-orbital sites. Vogl's sp^3s^* model adds one excited s^* orbital per site and only the declared nearest-neighbor s^*–p couplings, producing the maintained 10×10 block form

$$H(\mathbf{k}) = \begin{pmatrix} D_a & T(\mathbf{k}) \\ T(\mathbf{k})^\dagger & D_c \end{pmatrix}, \qquad D_b = \operatorname{diag}\big(E(s,b), E(p,b), E(p,b), E(p,b), E(s^*,b)\big). \tag{20}$$

The Table 1 transfer parameters determine $T(\mathbf{k})$; the s^*–s and s^*–s^* transfers are structural zeros in this model.

References

[1] P. Vogl, H. P. Hjalmarson, and J. D. Dow. A semi-empirical tight-binding theory of the electronic structure of semiconductors. *Journal of Physics and Chemistry of Solids*, 44(5):365–378, 1983.

[2] AiiDA Wannier90 Workflows contributors. aiida-wannier90-workflows. Software release 3.1.0, 2026. Pinned commit 1c0115cb7c77541a0bd549bc05005df435d651b6; moderate protocol and band-estimator source inspected 31 October 2026.

[3] Mahdisadjadi. Zinc-blende LCAO. Development Gist. Inspected 22 October 2026.

[4] Simon Phillpot. SiliconLCAO_4sDOS-v3, 2026. Development notebook; modifications recorded as 29 April–11 May 2026; inspected read-only 22 October 2026.

[5] Walter A. Harrison. *Electronic Structure and the Properties of Solids: The Physics of the Chemical Bond*. W. H. Freeman and Company, San Francisco, 1980.

[6] D. J. Chadi and M. L. Cohen. Tight-binding calculations of the valence bands of diamond and zincblende crystals. *physica status solidi (b)*, 68(1):405–419, March 1975.

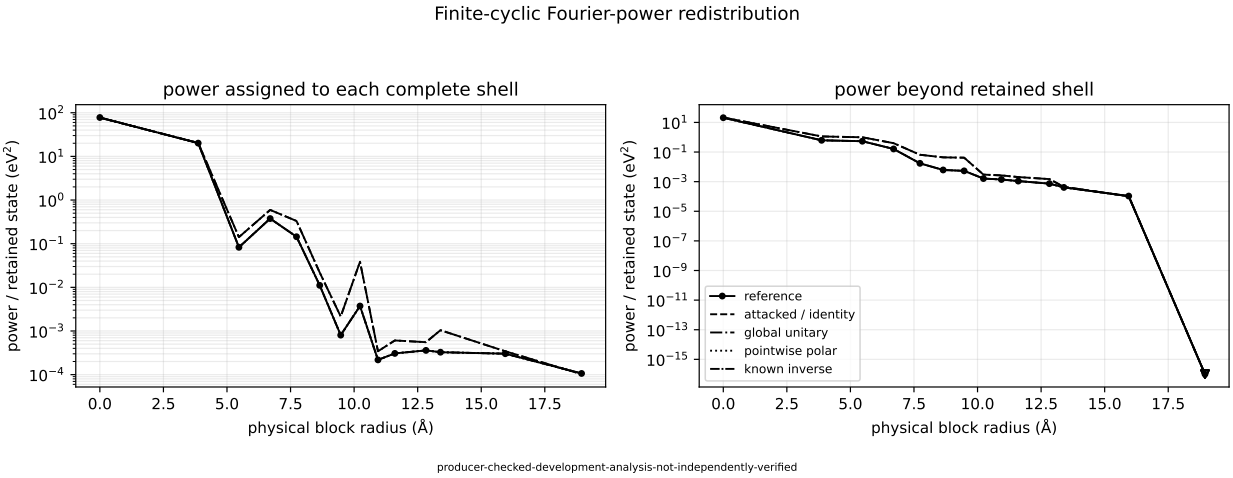


Figure 3: Reference and attacked block Frobenius power against physical block radius, together with cumulative retained and omitted power. Cyclic-to-ordinary removed power is displayed as a separate conversion channel. All numerical values are producer-checked development outputs and are not independently reconstructed.

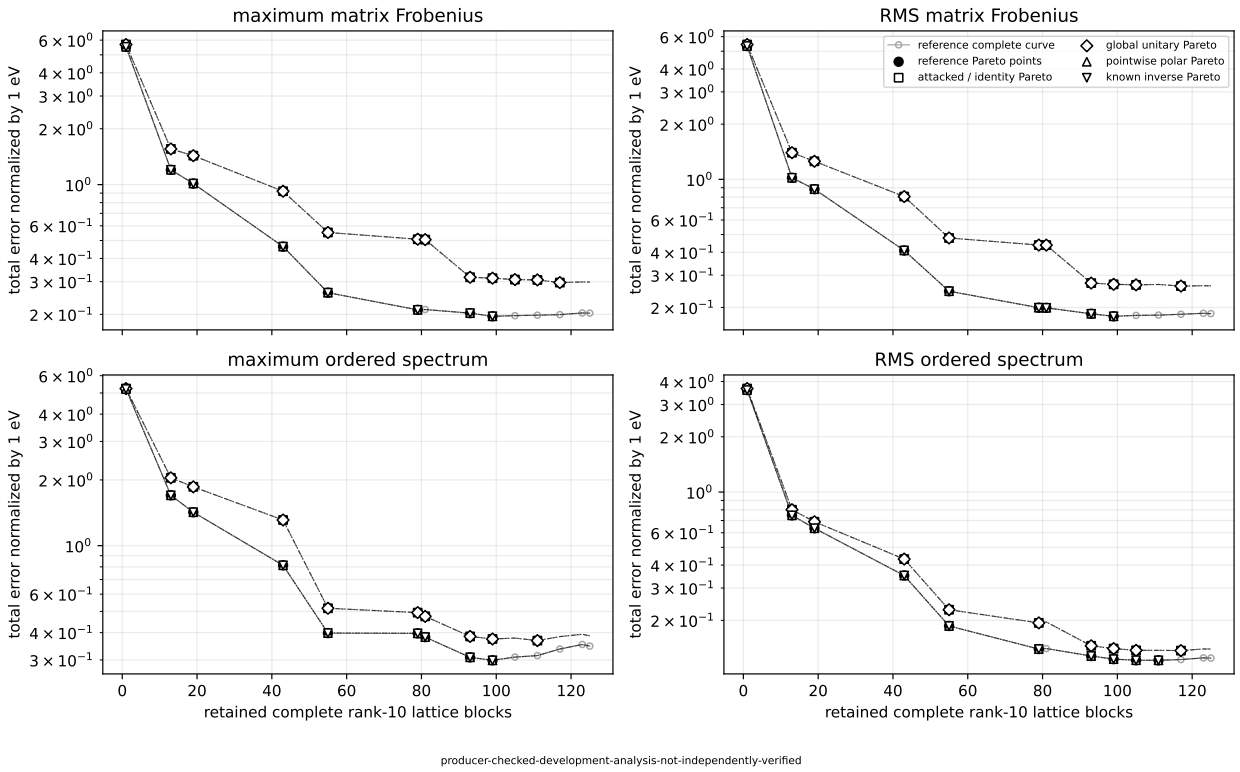


Figure 4: Complete normalized total-error curves against retained complete rank-10 block count for maximum matrix error, matrix RMS error, maximum sorted-band error, and sorted-band L2-RMS error. Emphasized markers identify the observed non-dominated supports for the reference and all four alignment channels; connecting segments are guides between the discrete shell inventory and are not interpolated support claims. All numerical values are producer-checked development outputs and are not independently reconstructed.

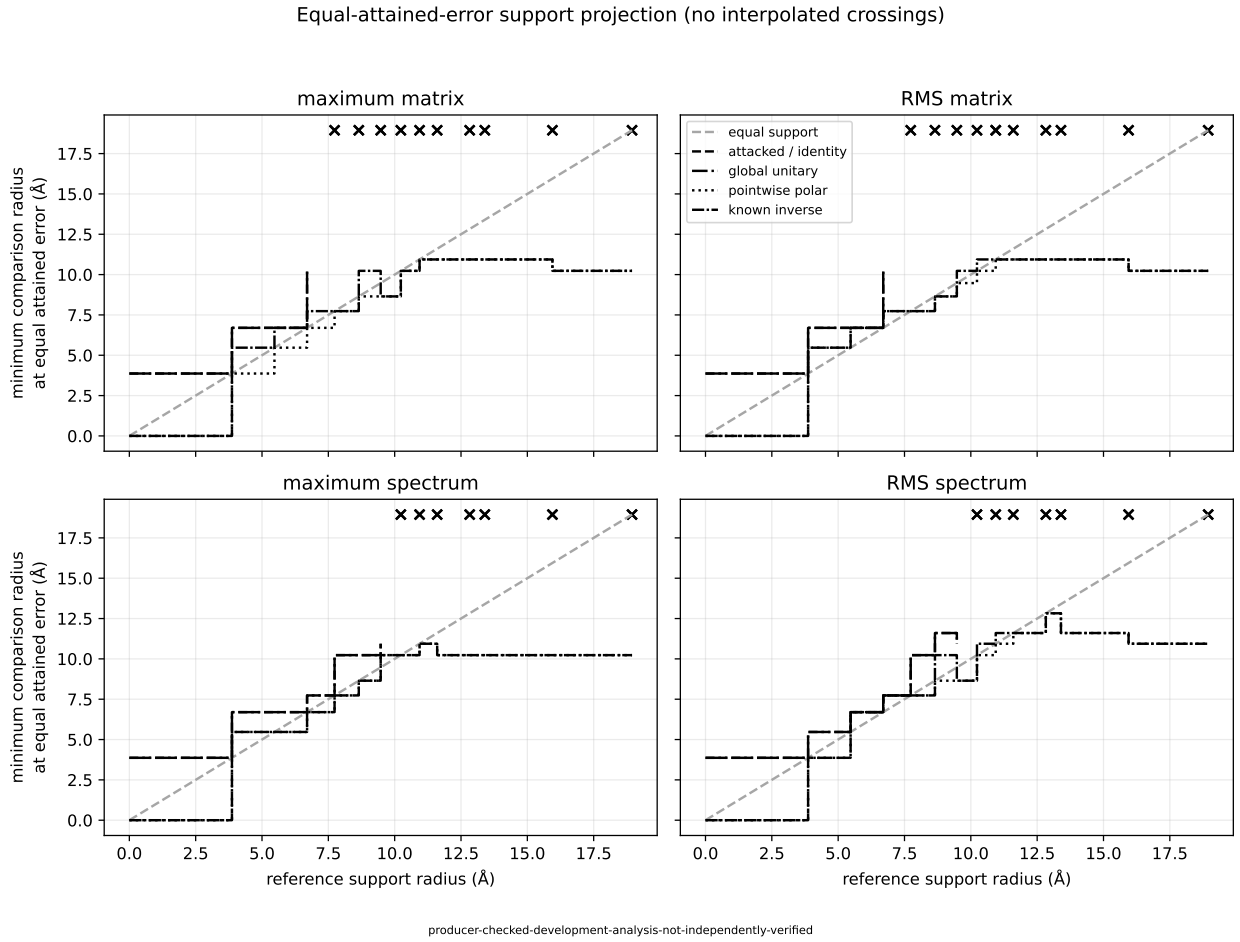


Figure 5: Empirical stair-step projection of the minimum comparison radius required at errors actually attained by each reference support. Crosses at the largest displayed radius mark errors not attained by the complete comparison inventory; no crossing is interpolated. All numerical values are producer-checked development outputs and are not independently reconstructed.

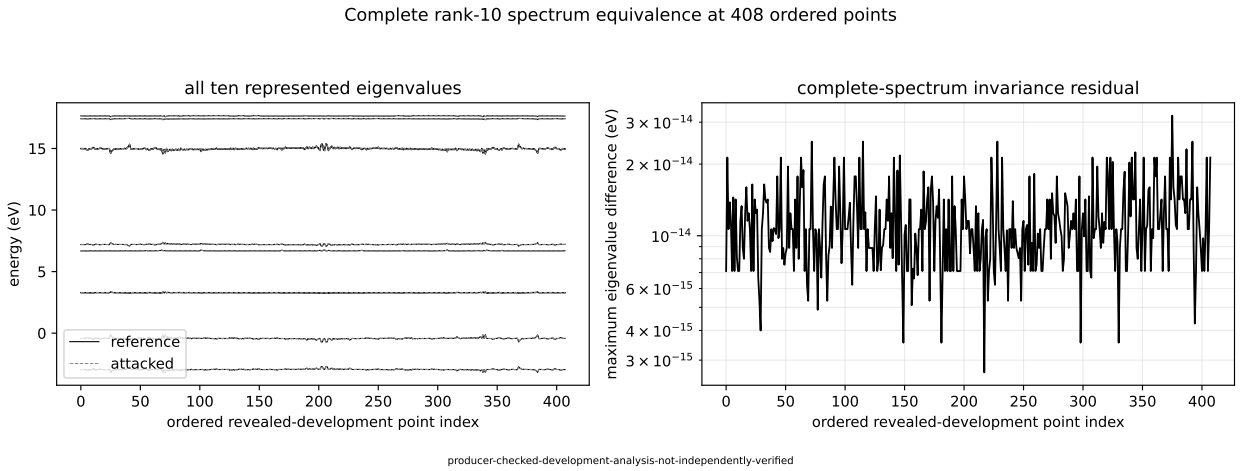


Figure 6: All ten reference and attacked eigenvalues and their pointwise maximum difference over the ordered 408-point development inventory. The point order is an inventory, not an interpolated reciprocal-space path. All numerical values are producer-checked development outputs and are not independently reconstructed.

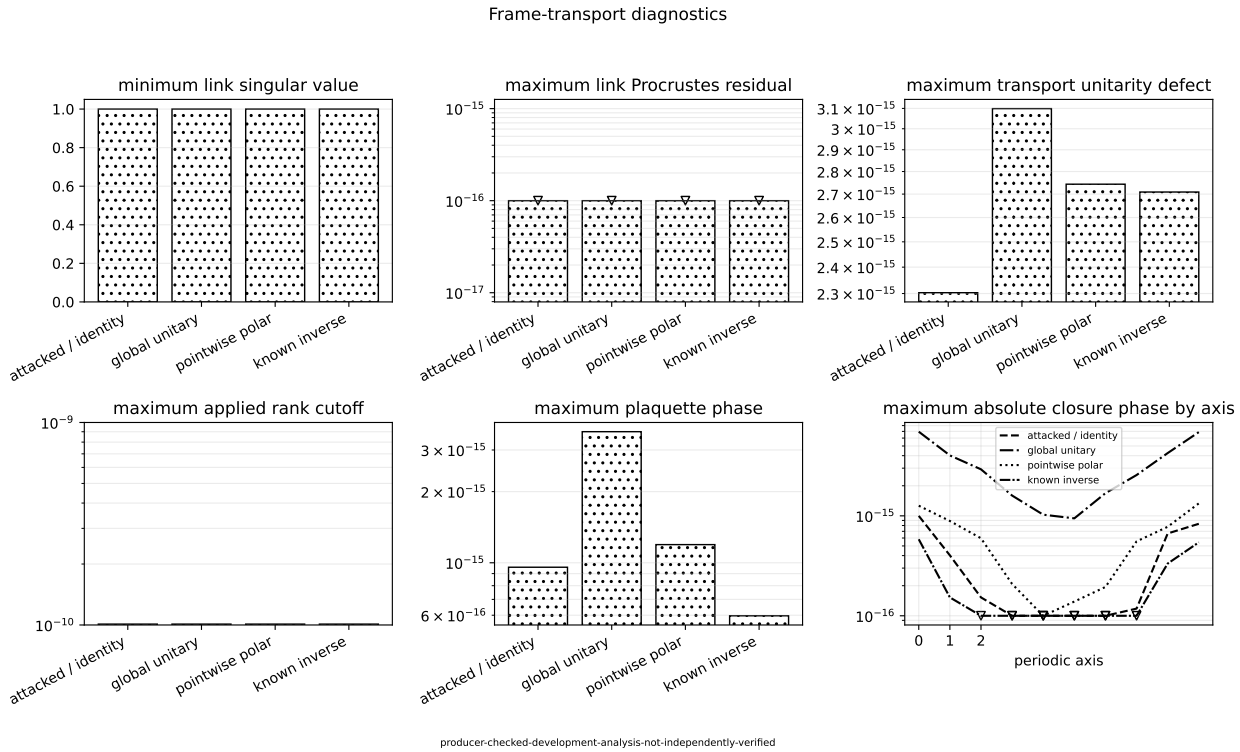


Figure 7: Minimum transport singular values, Procrustes residuals, unitarity defects, closure phases, and plaquette phases for all applicable alignment channels. All numerical values are producer-checked development outputs and are not independently reconstructed.

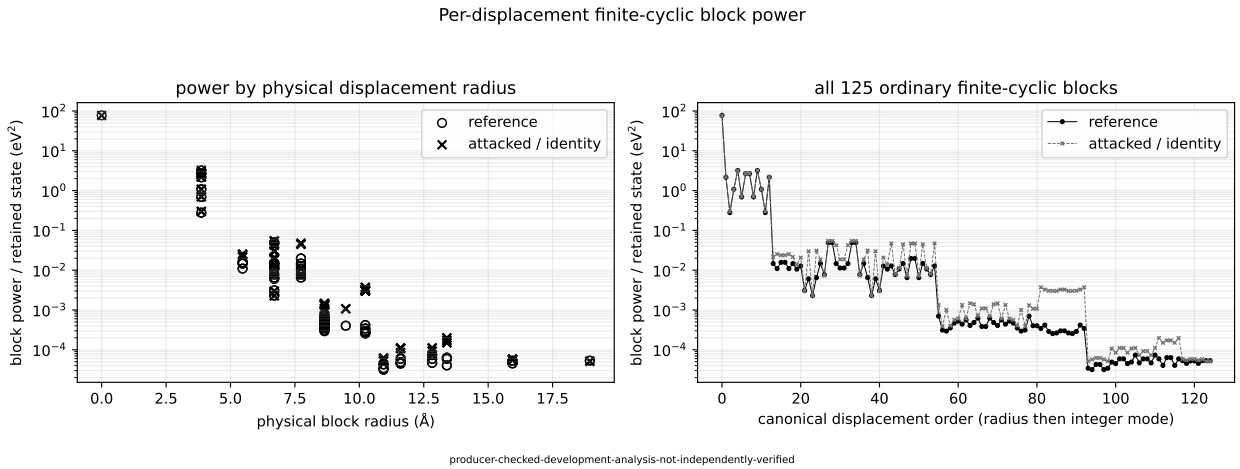


Figure 8: Per-displacement rank-10 block Frobenius powers for reference and attacked ordinary representations, indexed by displacement and physical block radius without assigning atomic-orbital or Slater–Koster semantics. All numerical values are producer-checked development outputs and are not independently reconstructed.

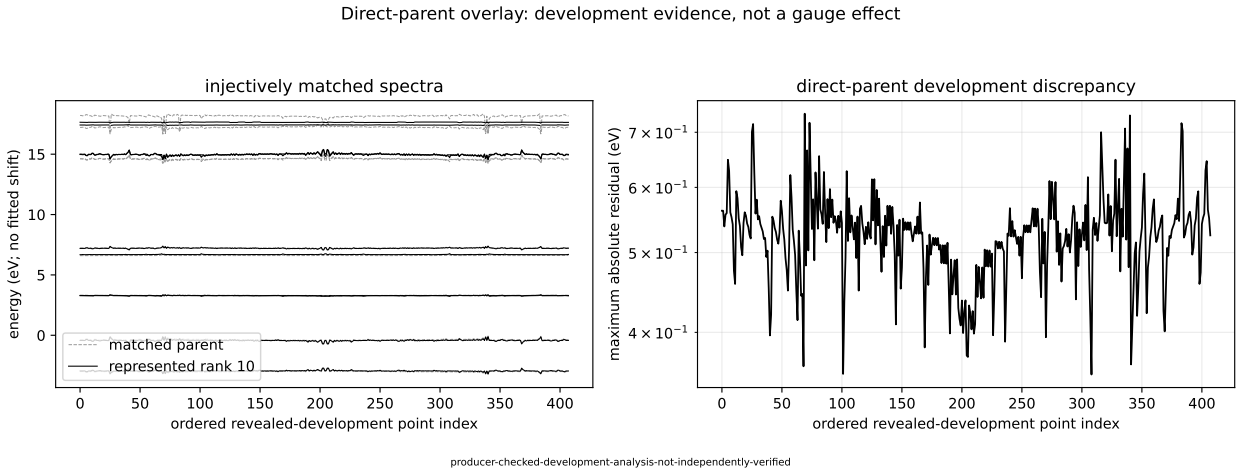


Figure 9: Separately labeled revealed-408 direct-parent band residuals by point role. This overlay is shown only after the internal represented-field curves are fixed and does not select the attack, support inventory, or any threshold. All numerical values are producer-checked development outputs and are not independently reconstructed.

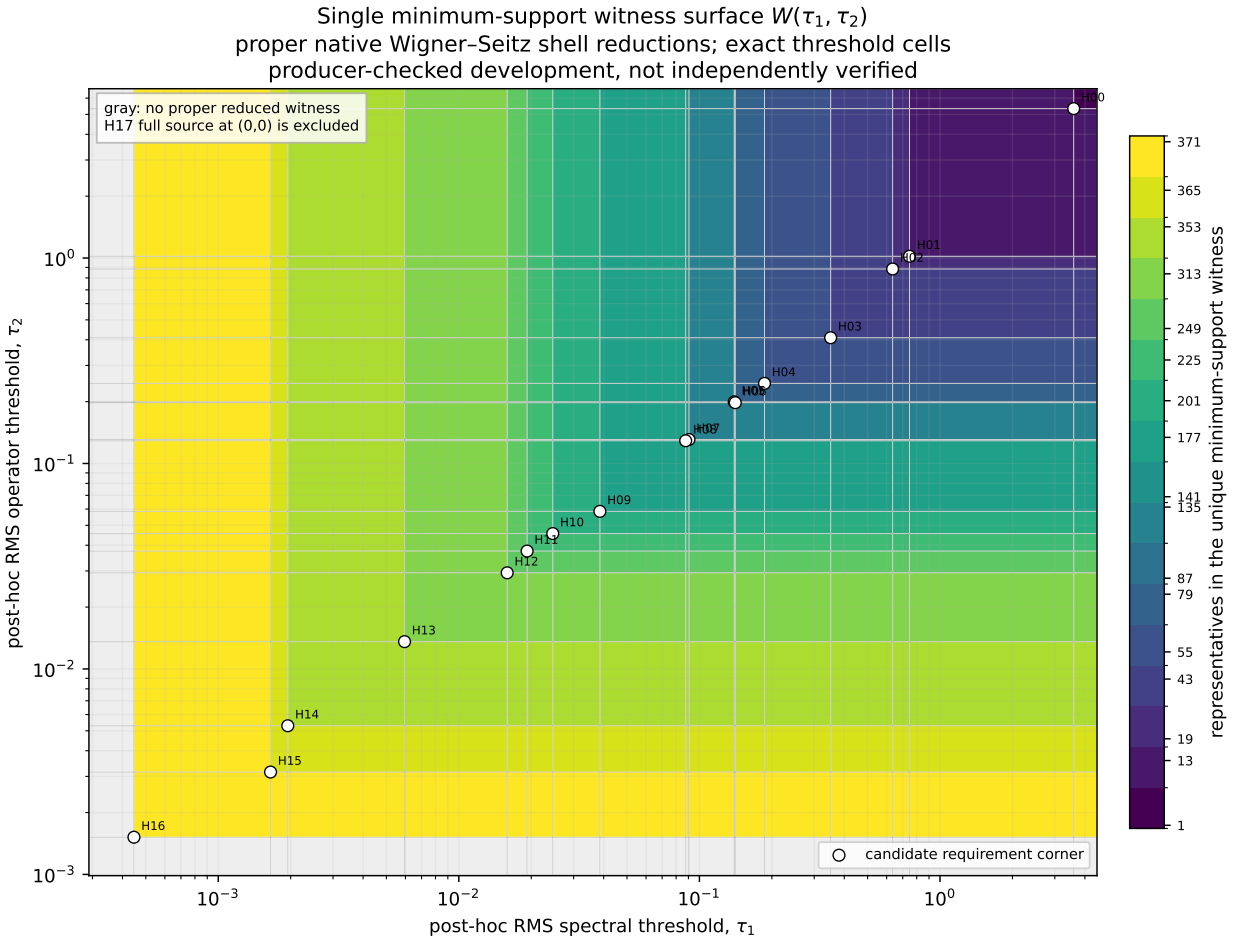


Figure 10: Each exact threshold cell is colored by the representative count of the unique minimum-support proper reduced shell satisfying both RMS inequalities. White circles mark candidate requirement corners. Gray cells admit no proper reduced witness; the exact full source at $(0, 0)$ is excluded. The cells are constructed from the observed requirement coordinates without interpolation or smoothing. All numerical values are producer-checked development outputs and are not independently reconstructed.

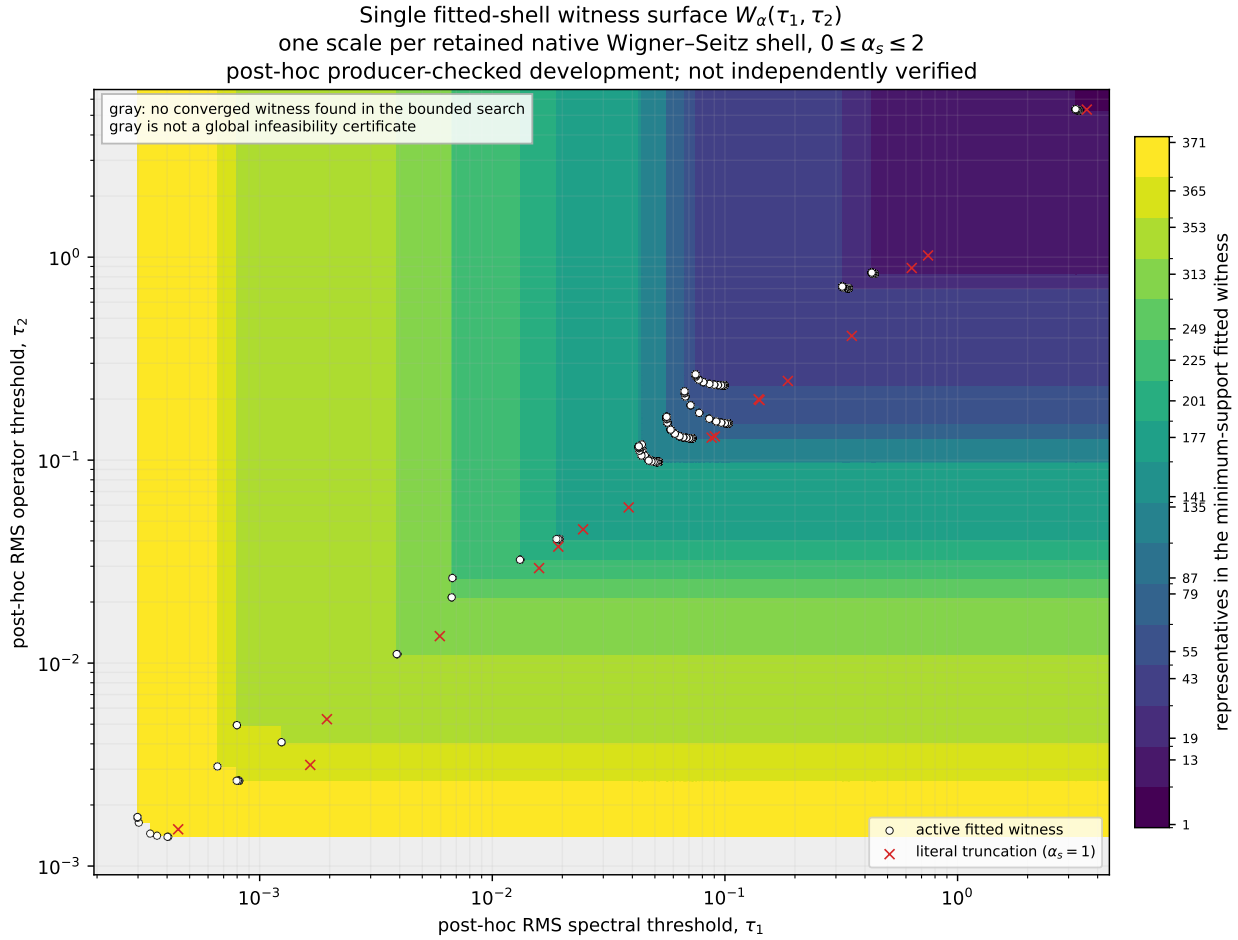


Figure 11: Each threshold cell is colored by the representative count of the minimum-support converged fitted witness found in the bounded search. White circles mark active fitted witnesses and red crosses mark literal $\alpha_s = 1$ truncations. Gray means that the search found no converged witness, not that infeasibility was certified. All numerical values are producer-checked development outputs and are not independently reconstructed.

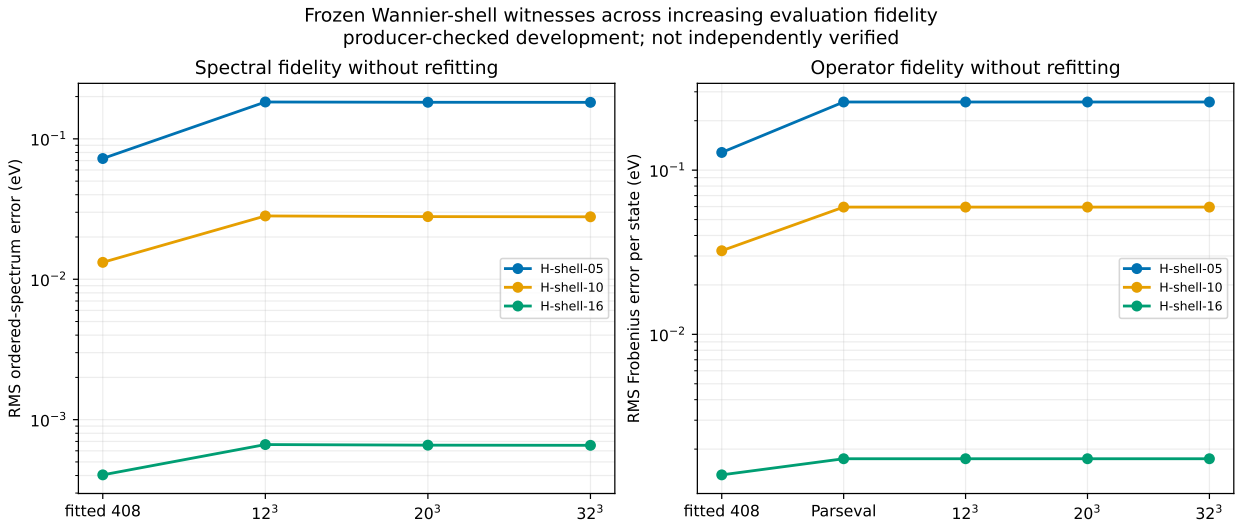


Figure 12: Left: fitted-408 and independent dense-mesh ordered-spectrum RMS errors. Right: fitted-408, exact Parseval, and sampled operator RMS errors. Parameters and supports are unchanged at every level. All numerical values are producer-checked development outputs and are not independently reconstructed.

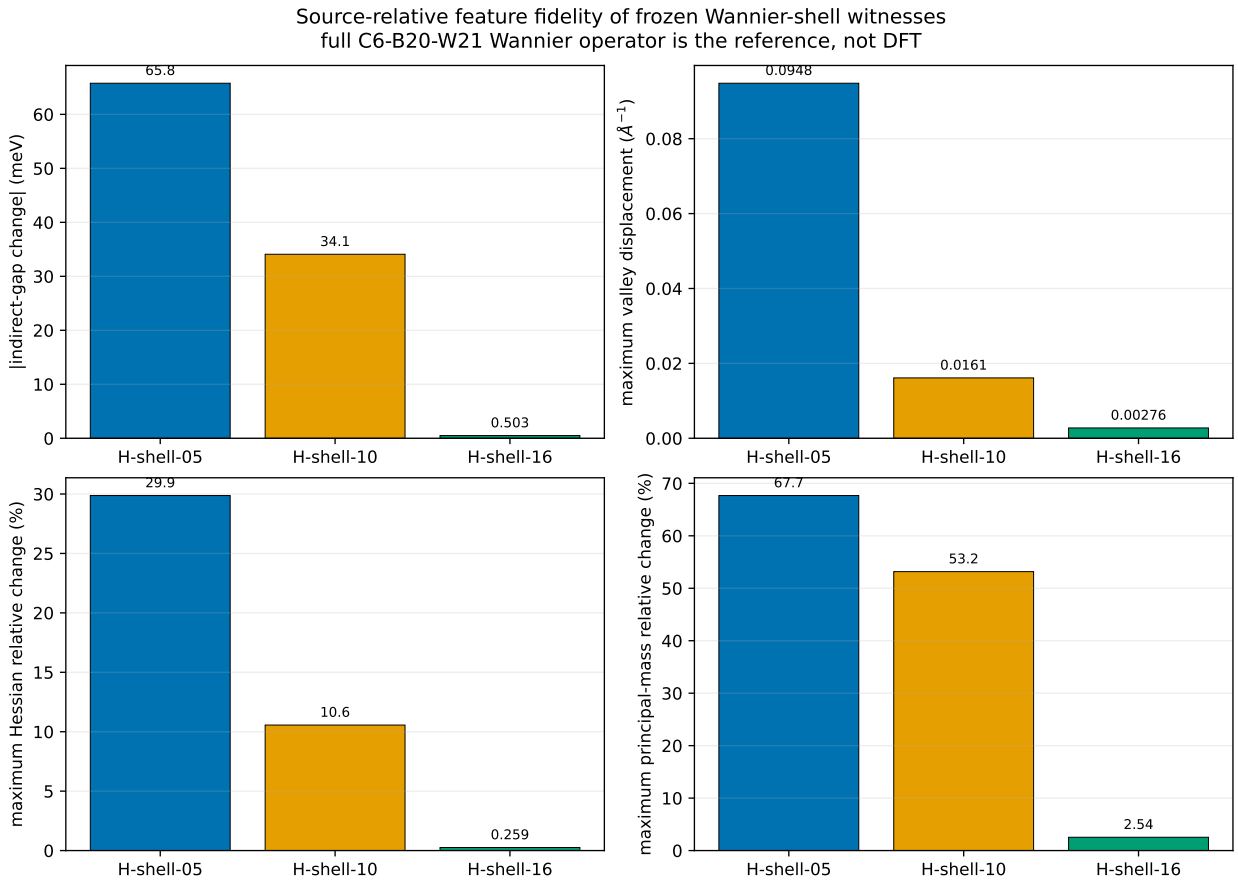


Figure 13: Absolute indirect-gap change, maximum valley displacement, maximum Hessian relative change, and maximum matched principal-mass relative change. The reference is the full C6-B20-W21 Wannier operator, not DFT. All numerical values are producer-checked development outputs and are not independently reconstructed.

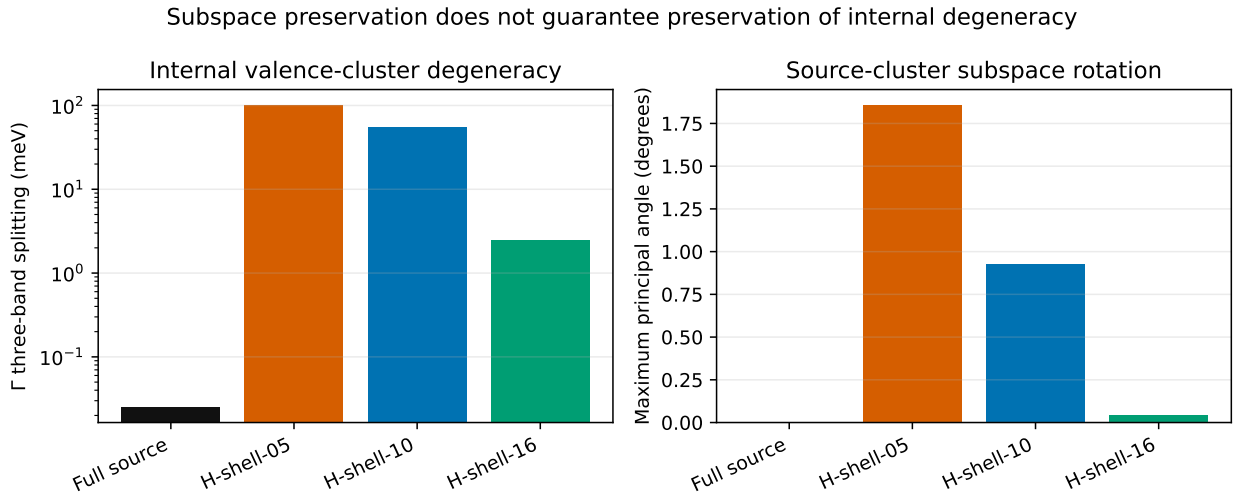


Figure 14: Three-band Gamma splitting and maximum source-cluster principal angle. The split and the subspace rotation are distinct error channels. All numerical values are producer-checked development outputs and are not independently reconstructed.

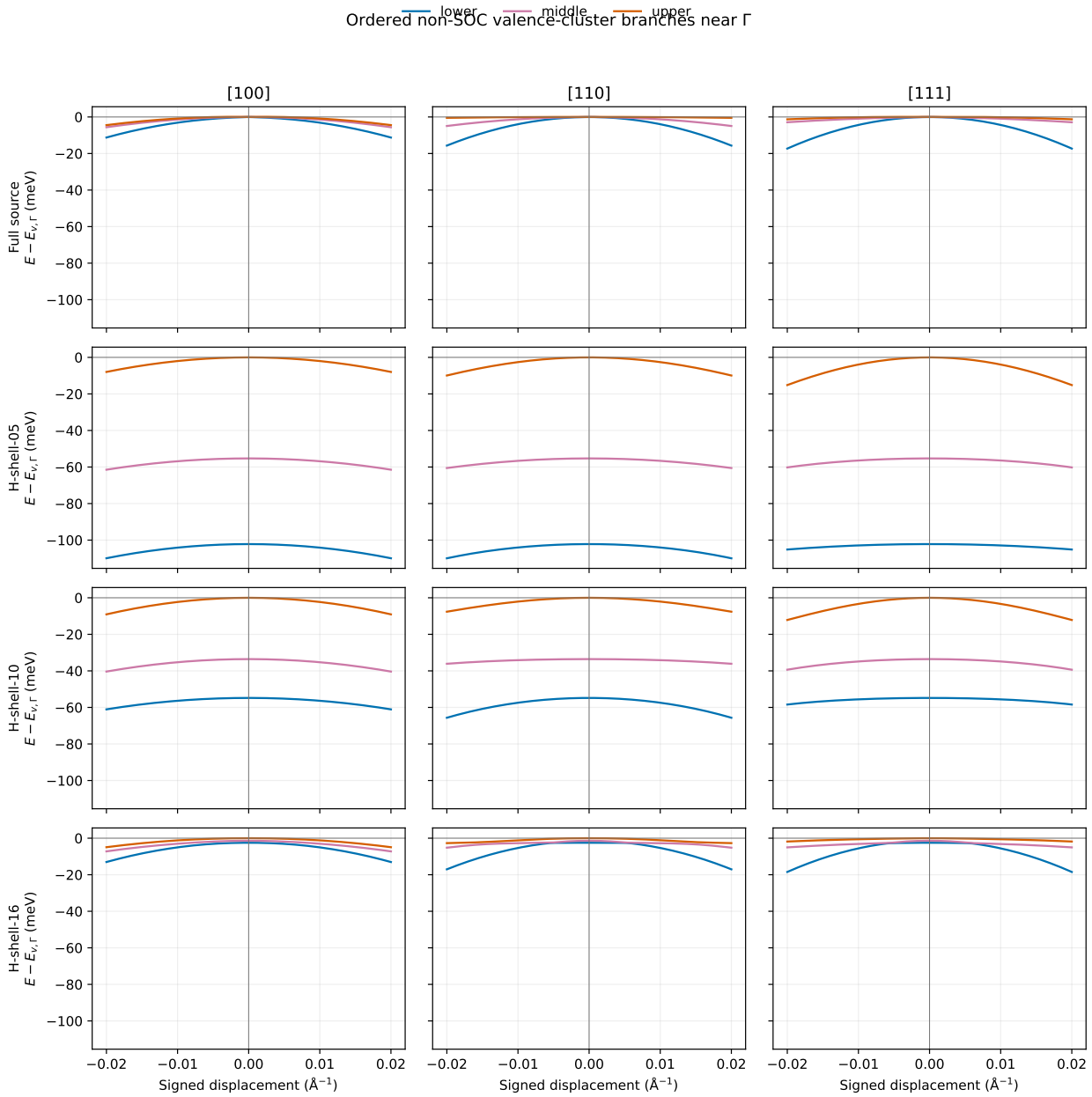


Figure 15: The three ordered cluster energies along [100], [110], and [111], referenced to each model's own Gamma VBM. Large vertical separation in H-shell-05 and H-shell-10 is the artificial internal cluster splitting, not a spin–orbit split-off band. All numerical values are producer-checked development outputs and are not independently reconstructed.

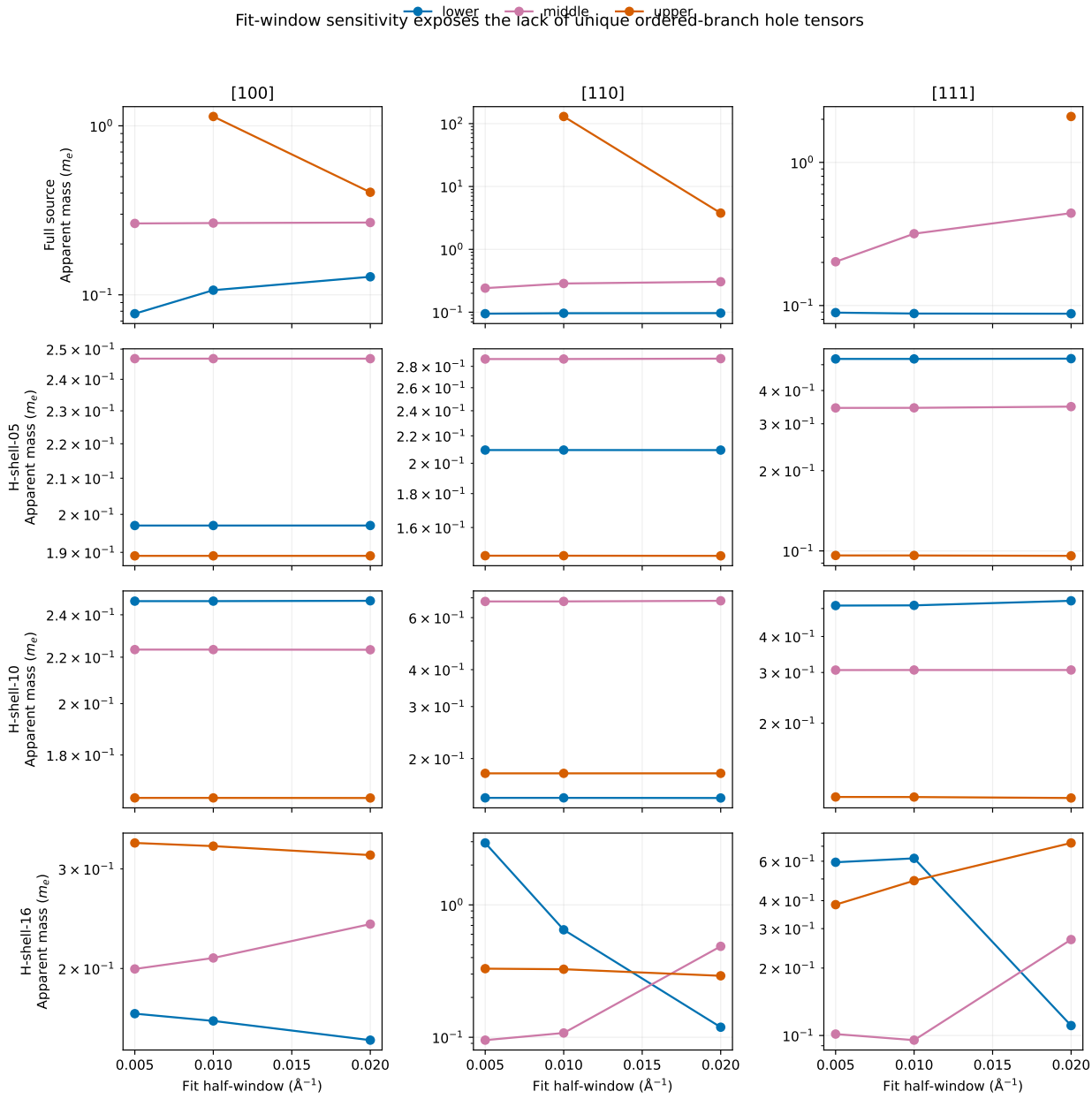


Figure 16: Apparent masses from independently ordered branch fits over three half-windows. Missing markers denote positive fitted curvature. The sensitivity demonstrates why these values are diagnostics rather than accepted hole tensors. All numerical values are producer-checked development outputs and are not independently reconstructed.

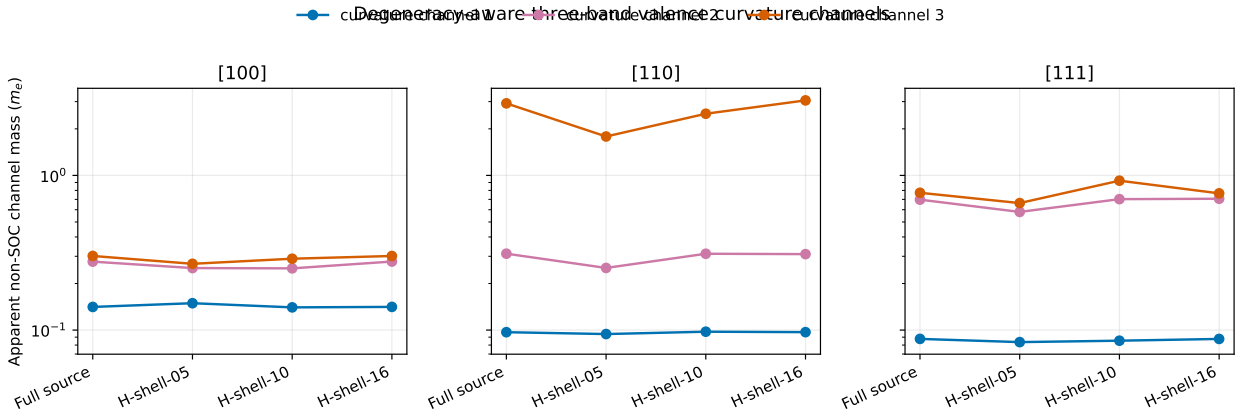


Figure 17: Three direction-contracted curvature-channel masses from the transported quadratic valence operator. They are multiband source-relative diagnostics and are not assigned physical HH, LH, or split-off labels. All numerical values are producer-checked development outputs and are not independently reconstructed.

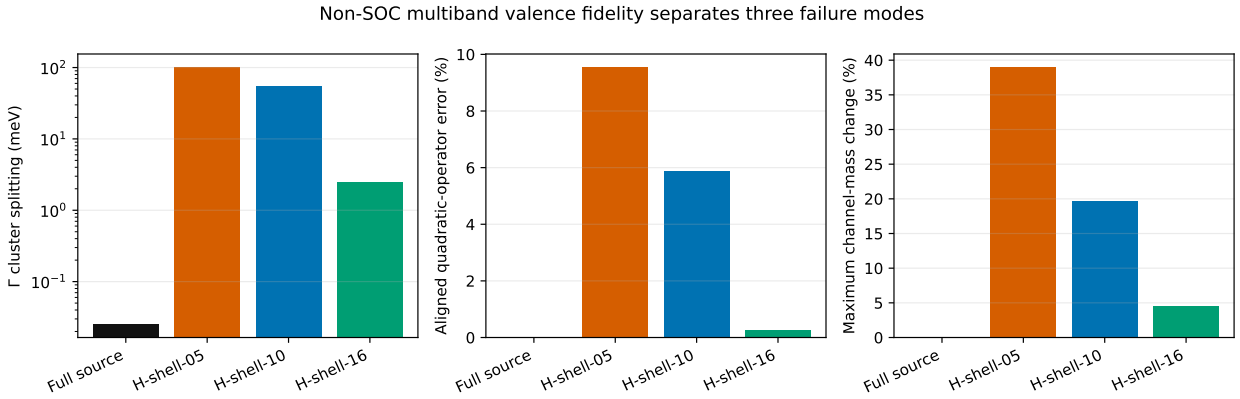


Figure 18: Gamma cluster splitting, aligned quadratic-operator error, and maximum directional curvature-channel mass change separate internal degeneracy, matrix-field, and derived curvature consequences. All numerical values are producer-checked development outputs and are not independently reconstructed.

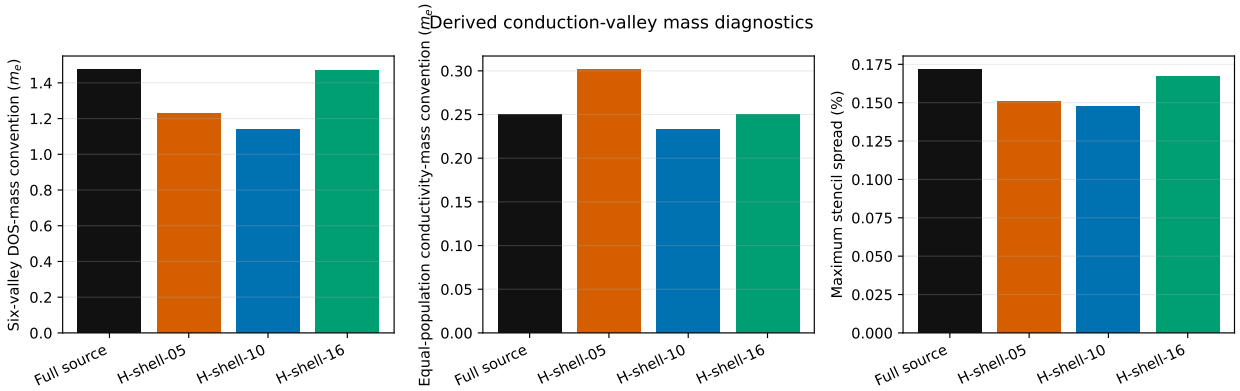


Figure 19: Six-valley DOS-mass and equal-population conductivity-mass conventions together with maximum finite-difference stencil sensitivity. These are curvature summaries, not transport calculations. All numerical values are producer-checked development outputs and are not independently reconstructed.

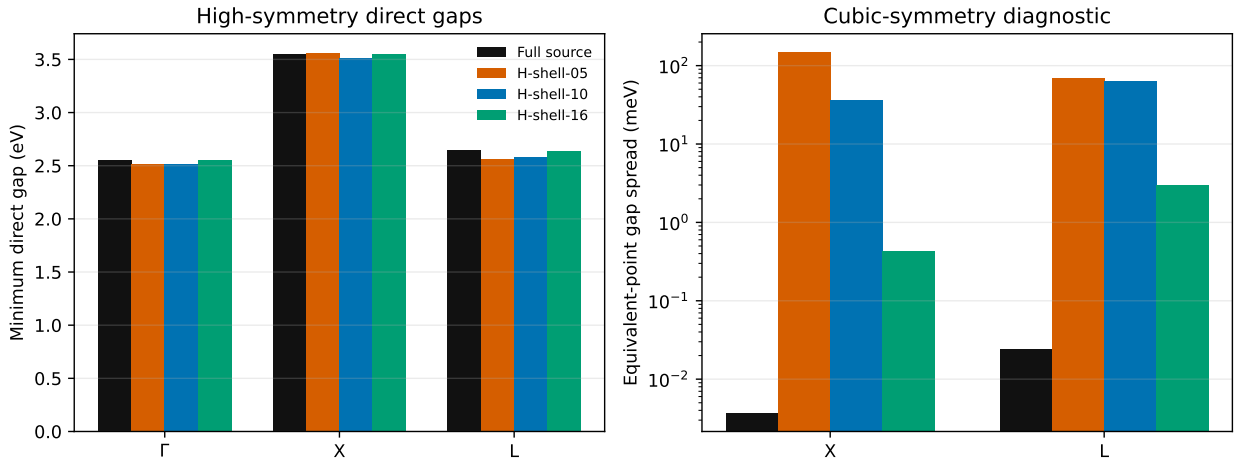


Figure 20: Minimum direct gaps at Gamma, X, and L and the spread among nominally equivalent X and L points. The spread is a represented-model symmetry diagnostic. All numerical values are producer-checked development outputs and are not independently reconstructed.